Business Intelligence

Project Handout

Spring Semester 2026

This handout details the rules for the Business Intelligence practical project.

First Delivery Deadline: March 22nd, 23h59

Second Delivery Deadline: May 10th, 23h59

Third Delivery Deadline: June 7th, 23h59

Project Group: 2 to 4 students

Project Summary

This project is meant to reinforce the conceptual knowledge that you will be acquiring throughout the course. In partnership with BI4ALL, which defined the case studies and data for the challenges, we ask you to apply core concepts of business intelligence tools and methodologies to deliver an end-to-end Self Service BI solution to support strategic, tactical, and operational business decisions in the scope of the challenges proposed.

During the first delivery of the practical project, each group is expected to select one challenge and to design a proper Dimensional Model that will be the basis to support the different levels of business decision makers of the chosen case study. For the second delivery, each group will implement the designed model in Fabric and develop an ETL Pipeline to load the Data Warehouse and Validate the loading process. Finally, in the third delivery the group will develop reports that answer the business questions and requirements of each challenge.

Group Rules

The project should be done in a group of between two (2) to four (4) students; we consider this the ideal size for group work, and do not allow other sizes.

Should you constitute groups of size other than stated size, you will be subjected to the following penalties: for each member in excess or below the stated size: penalty of **1.5 points** (out of 20 total points) in final project grade.

For example: should you choose to constitute a group of 6 members, your final project grade will be penalized in 3.0 points (out of a possible maximum grade of 20 points).

The only allowed exceptions that will be made regarding group size will be for situations beyond the control of the group members (for example, one of the students dropping out of the course) and will be evaluated on a case-by-case basis.

Source Data

The company BI4ALL has made available a set of three (3) challenges, each with their own data and business questions. Each group must choose one and only one challenge. The data sources provided in each challenge will be the foundation for the project.

The source data contains the following features:

i. Some sort of transactional records that can represent quantifiable facts, to be used in your future dimensional model; In terms of BI, transactional means data that is used to count, sum, or otherwise keep track of quantities of something that the organization deeply cares about.

ii. The quantifiable data that will make up the facts of your multidimensional solution represents several years of transactional history.

iii. Enough attributes (characteristics) so that you can extract at least 5 (five) dimensions. The characteristics you will set up as dimensions of your model should give you the ability to set up different hierarchies in different dimensions.

iv. The submitted solution, to be run by your teacher and BI4ALL Team, should address the defined requirements in the Project Scope file.

Project Deliverables and Solution Requirements:

1. Data Warehouse (First Delivery) – 5 points (out of 20):

Your group must deliver the following elements:

1. An initial **.pdf report,** containing the following points:
    i. Introduction and **Presentation of Business/Organization**.
    ii. Identification of **Business Problem**, including the context behind the informational problem that the BI system is going to solve.
    iii. Identification of the **Business Questions**. These should be clear and objective and will be the basis for the Dimensional Model and Power BI reports that you will be building.
    iv. Explanation of the content of the **source data** (no need for a data dictionary) and **discovery process**.
    v. A **design** of the **dimensional model** (star schema is recommended) and in-depth description of the model. The model should feature:
        a. At least **5 dimensions**, one of which must be a Time / Date dimension.
        b. At least **3 hierarchies**, with **average depth of 3 levels**.
        c. At least **one Fact table**, featuring at least **2 measures**.
        d. This design should include the **name of the tables** and **columns**, identify the **key columns** and **data types**, as well as the **relationships** between tables.
    vi. Explanation of the **data modelling methodology** followed (Kimball / Moody & Kortink) to achieve the Dimensional Model design.
    vii. Description of the **main components** (Dimensions, Facts) of the developed Dimensional Model.

2. A **Fabric Workspace** following the structure announced in Moodle on the delivery section and containing:
    i. **Creation of Lakehouse** – Setting up a Lakehouse and loading the necessary data for the analysis.

2. E.T.L. (Second Delivery) – 6 points + 1 (extra) point (out of 20):

Your group must deliver the following elements:

1. A **.pdf report,** containing the description of the following points:
    i. Integration of the contents of the previous delivery
    ii. Presentation and description of the **Data Warehouse**
    iii. Explanation of the **set up used for the ETL** (Pipelines, Dataflows and Notebooks)
    iv. Explanation and justification of the **Transformation steps applied**.

2. The same **Fabric Workspace** from the first delivery, plus:
   i. **Warehouse** – Fully loaded after the ETL process ran successfully.
   ii. **Pipeline** – The pipeline use to orchestrate de ETL.
   iii. The **Dataflows** created to transform and load the data into the Warehouse.
   iv. Use of **at least one Notebook** applying **advanced analytics or other predictive methods.** The output should be a new table/column(s) to be integrated in the Warehouse.
   v. As much as possible, the ETL processes should employ a varied set of techniques and transformations, reflecting the different aspects learned in the laboratory sessions. (ex: Merge, Conditional Column, Group By, etc.)
3. (OPTONAL) Complex ETL developments not taught in class will count as extra points:
   i. **Incremental Load** – Successful ETL ran uploading only new data.
   ii. **Pipeline Activities** – Using complex activities in the Pipeline environment for ETL purposes.
   iii. **Transformations** – Using complex M/SQL/Pyspark transformations in a dataflow/notebook.

3. Reporting (Final Delivery) – 7 points + 1 (extra) point (out of 20):

Your group must deliver the following elements:

1. A final **.pdf report,** containing the description of the following points:
   i. Integration of the contents of the previous delivery.
   ii. Presentation and description of the **Semantic Model** (hierarchies, hidden columns, data types, formatting, summarization, date table, etc.)
   iii. **Explanation of the measures** created and how these are important to answer the business needs.
   iv. Presentation and Description of the **Reports created**.
   v. Explanation of the **main technical aspects of the report pages** in terms of the visualization techniques.
   vi. **Analysis and discussion** of the outputs of the project. Please be precise in explaining **how the solution answers the questions from the 1st Report**.
   vi. Conclusion, summarizing the main aspects of the project and key takeaways.

2. A **Fabric Workspace** containing:
   i. **Semantic Model** - hierarchies correctly configured, column data types, sort, summarization, date table, etc.).
   ii. At least **5 (five) DAX Measures,** of which at least **2 (two)** should be a **time intelligence measure** (make sure these are able to provide the basis to answer the business needs).
   iii. One **Fabric Paginated Report** table visualization.
   iv. One **Power BI report** with:

a. at least 3 pages, each page with a minimum amount of 4 visualizations (slicers and other filter-like visualizations can and should be used, but will not count as one visualization).

b. At least **one KPI (Key Performance Indicators) visuals** for business monitoring.

c. At least **one visual** integration machine learning and /or predictive analytics results that answers a Business Question.

v. Works that **can respond to all business needs** following **data visualizations best practices** will be valued.

3. (OPTONAL) Complex Reporting developments not taught in class will count as extra points:

iv. **Complex interactions** – Successful use of filters/slicers to achieve unique perspectives.

v. **Original Visuals** – Using non-standard visuals that answer a BQ in a better way.

vi. **Calculated Measures** – Using complex DAX formulas.

Delivery Guide

When submitting the group results, please make sure to follow the following instructions:

4. One member of the group should deliver through the Moodle Project delivery section a report in the **.pdf** format.
5. This report should contain a **title page** which should include the **group's number** and the **name and student number** of each member.
6. The **report name** should be **Report XX** (where XX stands for the group's number).
7. The Fabric Workspace name should be **BI MAA 2025 Group XX.**
8. One member of the group will have to **give Administrator Access to the Fabric Workspace** to the Lab professor at the Delivery Date for each delivery.

Final Notice:

- **failure to deliver on time will incur a 0.5 point penalty for each late day** (for example, 4 late days will accrue a 4.0 point penalty).
- **Failure to comply with the delivery guide** (no proper naming of objects, duplicated or unclear files, improper folder configuration, etc.) will meet with a **0.5 point penalty**.
- **For the second and third deliveries, extra work will account for 10% of the grade (up to 4 different extra developments).** We consider extra work any **developments that were not shown in class, that show a high level of complexity and make sense in the context of the solution**. **The extra work must be stated in the report as such**.

Questions and Clarifications

Should it be necessary, do not hesitate to seek help from your Lab teacher, or post a question on the Moodle project forum.

Good luck with your project!